

Lab Worksheet - 7

Verilog programming

CSD 211 computer organization & architecture

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Question Implement a 4 bit multiplier in HDL verilog, using full adders.

Hint- Store the intermediate results to use them later to perform complete multiplication.

Write a design and testbench, both the codes for the above given problem to simulate the result.

Example-

How multiplication is performed?

1011	(this is 11 in decimal)
1110	(this is 14 in decimal)
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0000	(this is 1011 x 0)
1011x	(this is 1011 x 1, shifted one position to the left)
1011xx	(this is 1011 x 1, shifted two positions to the left)
+ 1011xxx	(this is 1011 x 1, shifted three positions to the left)
=====	
10011010	(this is 154 in decimal)